Milky Way foreground-screen stratification in a SPARC/HECATE residual audit

Abstract

The Milky Way foreground is an observational layer, not merely a nuisance. Zone of Avoidance work with JWST/NIRCam has recently shown that faint background galaxies can be recovered through heavily contaminated Galactic regions, while also emphasizing that extinction, stellar crowding, and confusion noise shape extragalactic visibility. Motivated by this, we define a deliberately simple foreground-screen diagnostic for the public SPARC residual-disturbance audit: Galactic latitude computed from HECATE RA/DEC for exact-name SPARC overlaps. Using the provisional screen `|b| <= 24 deg`, the packet selects 18 systems. In the reviewed A/C subset, this screen is C-dominant: A=1 and C=6, with 11 ambiguous B systems. The A/C enrichment check is not statistically decisive (`p = 0.295343139746`), so the result is best
read as an observer-screen audit and preregistration target, not a detection.

Motivation

Nilo-Castellon et al. identify 102 galaxies in a JWST/NIRCam field in the Zone of Avoidance and describe Galactic extinction, stellar crowding, and confusion noise as historical limits on background-galaxy detection. This motivates a conservative question for SPARC residual audits: do systems projected nearer to the Milky Way plane occupy a visibly different label/observability stratum?

The point of this note is not to explain galaxy dynamics. It is to preserve a small, reproducible foreground-screen packet that can be tested later with better extinction maps, star-count fields, and independent galaxy samples.

Data And Construction

Inputs:

- HECATE v1.1 RA/DEC for exact-name SPARC overlaps.
- Residual-blind A/B/C labels from the public Paper 1 SPARC residual-disturbance packet.

- Optional residual summary fields from the public disturbance-inference packet.

The script converts HECATE equatorial coordinates to Galactic longitude and latitude using the standard J2000 north Galactic pole constants. The primary screen is:

low_latitude_screen = |b| <= 24 deg

This threshold is not claimed as fundamental. It is a transparent reconstruction of the remembered 18-object screen and should be stress-tested in future work.

Main Diagnostic

```
| Metric | Value | Interpretation |
| --- | --- | --- |
| hecate_sparc_overlap_rows | 120 | exact-name SPARC/HECATE overlaps with RA/DEC |
| low_latitude_threshold_deg | 24.0 | preliminary foreground-screen proxy |
| low_latitude_total | 18 | | b | <= 24 deg |
| low_latitude_A | 1 | reviewed calm/regular class count |
| low_latitude_B | 11 | ambiguous class count |
| low_latitude_C | 6 | reviewed disturbed class count |
| high_latitude_A | 15 | | b | > 24 deg |
| high_latitude_B | 57 | | b | > 24 deg |
| high_latitude_C | 30 | | b | > 24 deg |
| ac_one_sided_c_enrichment_p | 0.295343139746 | diagnostic only; not a detection |
```

Threshold Scan

```
| AbsGalacticLatitudeThresholdDeg | N_total | N_AC | N_A | N_B | N_C |
| --- | --- | --- | --- | --- | --- |
| 5 | 0 | 0 | 0 | 0 | 0 |
| 7.5 | 0 | 0 | 0 | 0 | 0 |
| 10 | 1 | 0 | 0 | 1 | 0 |
| 12 | 2 | 0 | 0 | 2 | 0 |
| 15 | 8 | 2 | 0 | 6 | 2 |
| 20 | 13 | 5 | 0 | 8 | 5 |
| 22 | 15 | 6 | 1 | 9 | 5 |
| 23 | 17 | 6 | 1 | 11 | 5 |
| 24 | 18 | 7 | 1 | 11 | 6 |
| 24.5 | 18 | 7 | 1 | 11 | 6 |
| 25 | 19 | 8 | 1 | 11 | 7 |
| 26 | 20 | 9 | 1 | 11 | 8 |
| 27 | 21 | 10 | 2 | 11 | 8 |
| 28 | 21 | 10 | 2 | 11 | 8 |
| 29 | 23 | 12 | 2 | 11 | 10 |
| 30 | 25 | 13 | 3 | 12 | 10 |
```

The 18 Low-Latitude Systems

```
| GalaxyName | Class | AbsGalacticLatitudeDeg | GalacticLongitudeDeg |
HecateDistanceMpc | Projection_RMS | ResidualDisturbanceScore_v01 |
| --- | --- | --- | --- | --- | --- | --- |
| UGC03205 | B | 8.206377 | 342.903037 | 54.37 | | |
| NGC6946 | B | 11.672188 | 60.144887 | 5.516 | | |
| UGC11557 | B | 12.814719 | 60.883601 | 23.66 | | |
| UGC00731 | C | 13.149452 | 29.723502 | 13.03 | | |
| NGC6674 | B | 13.919223 | 101.271460 | 54.11 | | |
```

```
| UGC02885 | B | 14.050142 | 356.283096 | 71.12 | | |
| UGC02916 | C | 14.194665 | 19.057928 | 66.11 | | |
| ESO563-G021 | B | 14.397417 | 270.996728 | 79.44 | | | |
| NGC0891 | C | 17.415249 | 15.479606 | 9.111 | | |
| NGC1003 | B | 17.545110 | 11.859346 | 10.76 | | |
| NGC2915 | C | 18.357173 | 223.897427 | 4.03 | | |
| UGC12632 | C | 19.311054 | 49.092790 | 9.204 | | |
| UGC02259 | B | 19.796208 | 8.718768 | 9.954 | | |
| NGC7331 | A | 20.724275 | 62.141728 | 14.4 | | |
| NGC6789 | B | 21.517771 | 60.889695 | 3.267 | | |
| NGC0801 | B | 22.456867 | 17.644924 | 55.53 | | |
| UGC11455 | B | 22.844922 | 52.115383 | 85.75 | | |
| NGC3109 | C | 23.070239 | 253.762265 | 1.291 | | |
```

Interpretation

The result is directionally interesting because the low-latitude screen is C-heavy among systems that already have reviewed A/C labels. However, the sample is small and B-dominated. A one-sided A/C hypergeometric enrichment diagnostic returns `p = 0.295343139746`, which is not a discovery-level result.

The correct reading is therefore:

The Milky Way foreground screen is a real observability stratum. In this SPARC/HECATE packet, a reconstructed 18-object low-|b| screen is C-dominant among reviewed A/C labels, but the result remains a preregistered audit target rather than a physical inference.

Next Tests

1. Replace the latitude-only proxy with `E(B-V)`, `A_V`, star-count density, and local confusion metrics.
2. Freeze the screen thresholds before residual inspection.
3. Re-run the audit on independent low-latitude and high-latitude galaxy samples.
4. Separate foreground observability effects from intrinsic disturbance evidence.
5. Report all A/B/C transitions under threshold scans rather than selecting only the most favorable boundary.

Claim Boundary

This note does not claim new dynamics, a physical detection, or proof of any private parent theory. It is a public observer-screen method note.

References

- J. L. Nilo-Castellon et al., *Faint galaxies in the Zone of Avoidance revealed by JWST/NIRCam*, arXiv:2510.12488, A&A; 704, A209 (2025).
- SPARC residual-disturbance Paper 1 public reproducibility package.
- HECATE v1.1 public catalog.